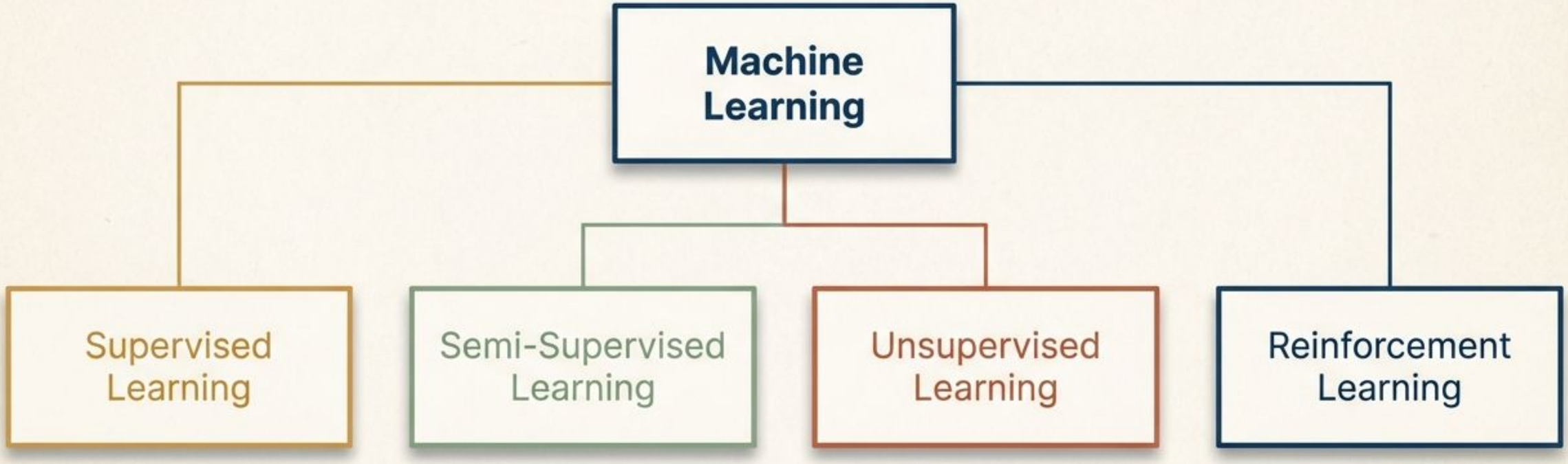
Mapping the Machine Learning Landscape

A visual guide to how algorithms learn, categorized by the core currency of AI: human supervision.



Algorithms require data to learn, but they differ fundamentally in how much human guidance they need to succeed.

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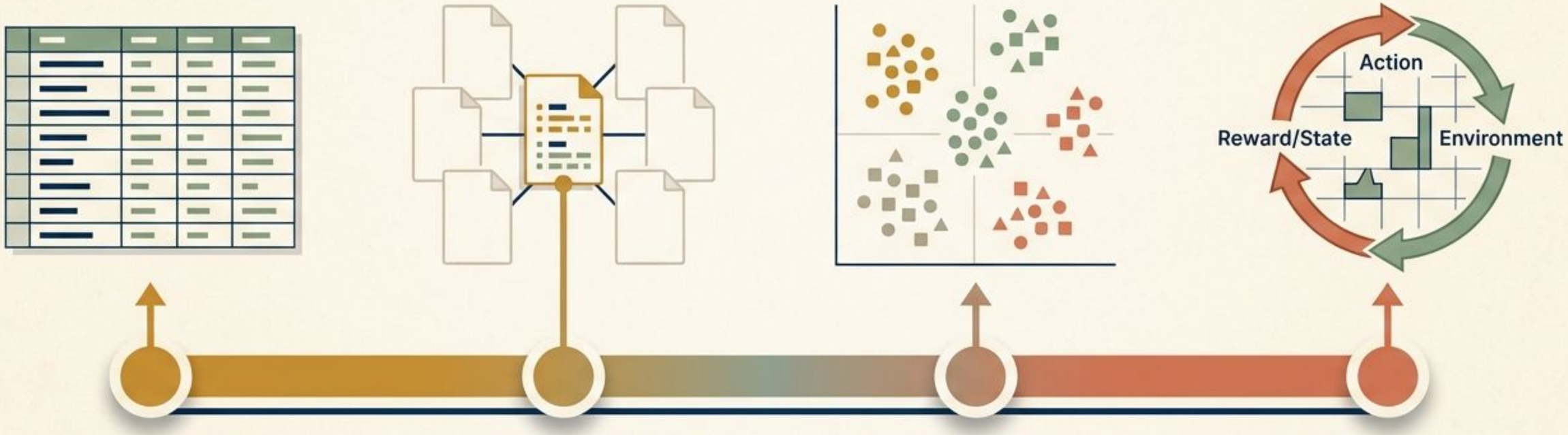
The Supervision Spectrum

The four primary categories of machine learning architecture are dictated entirely by the amount of human labeling required.

High Human Input / Cost

Zero Human Input / Environment-Based



Supervised
(100% Labeled Data)

Semi-Supervised
(Hybrid Data)

Unsupervised
(0% Labeled Data)

Reinforcement
(Zero Static Data)

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Supervised Learning: Learning by Example

The Inputs	
IQ	CGPA
87	7.1
111	8.9
75	6.3

Algorithmic Transformation

The Known Output (Label)
Target: Placement
Yes
Yes
No

Key Insight: The algorithm's sole job is to discover the mathematical relationship between 5,000 rows of known inputs and known outputs, allowing it to predict outcomes for future, unseen students.

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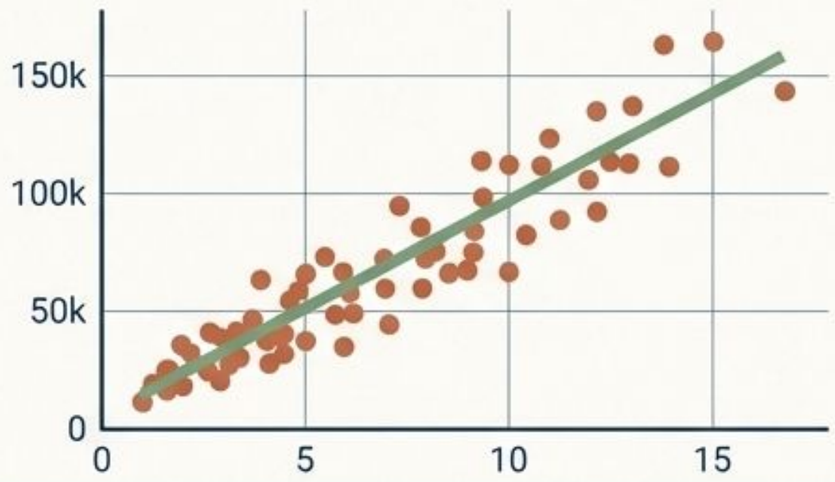
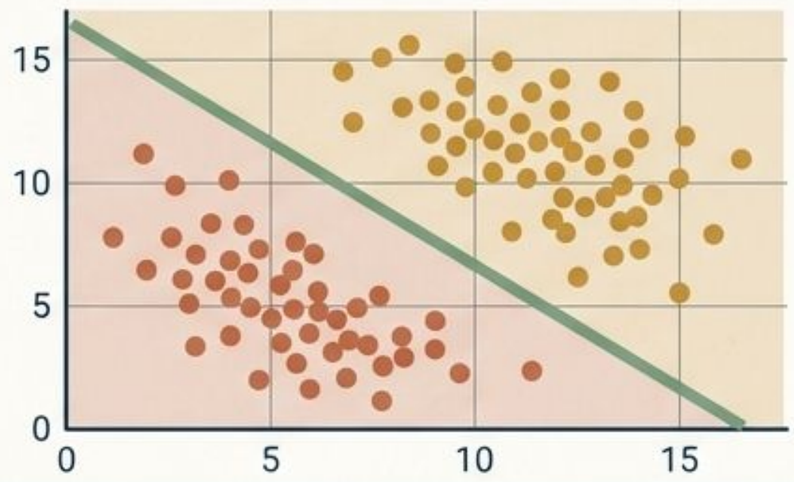
The Two Branches of Supervised Learning

Regression		Classification	
	How much?		Which one?
	Outputs a Continuous Numerical Value		Outputs a Categorical Label
Visual Model		**Visual Model**	
Real-World Example	Predicting a graduate's exact starting salary in dollars based on CGPA.	**Real-World Example**	Predicting whether an email is Spam or Not Spam based on keywords.

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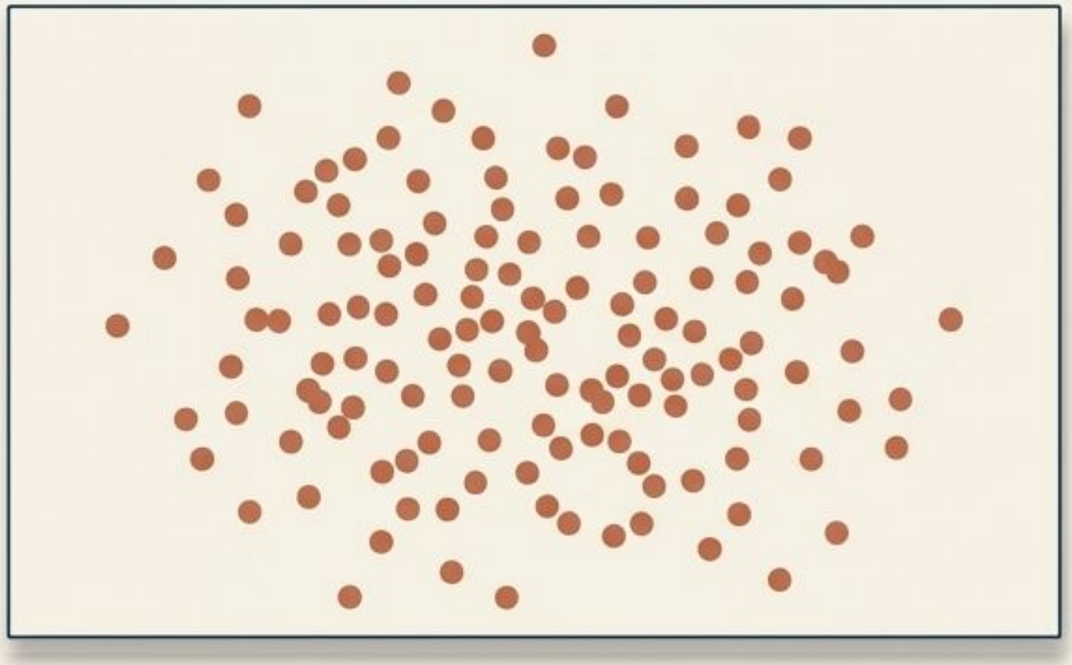
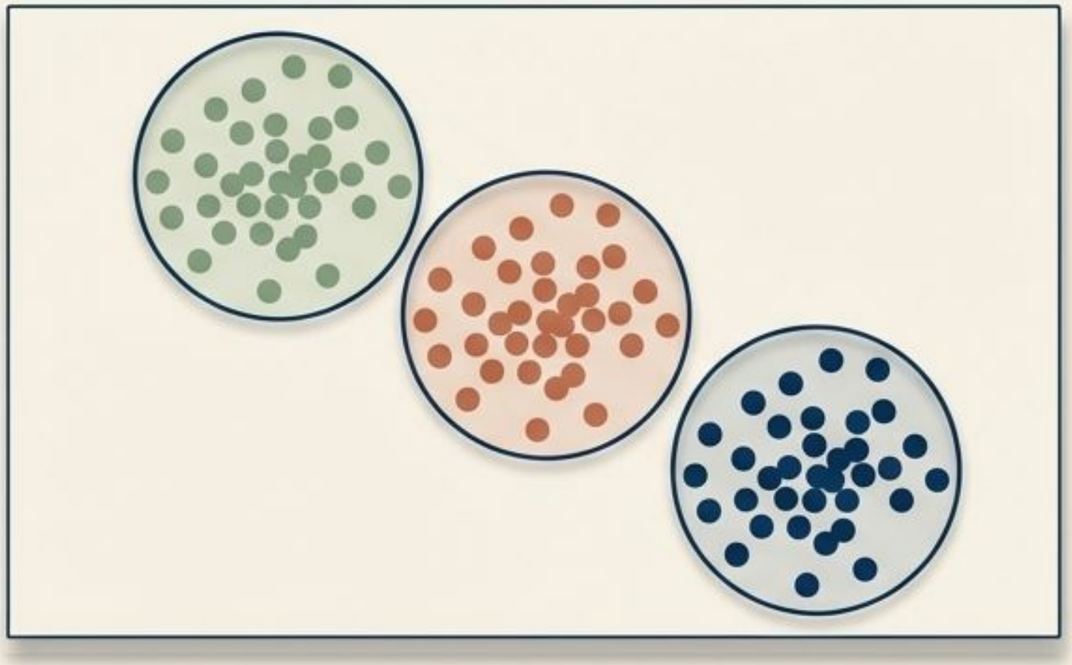
Unsupervised Learning: Discovering Hidden Structure

The Raw Input



Only inputs provided (e.g., IQ, CGPA).
No labels, no known outcomes.
Directed prediction is impossible.

The Machine's View



The algorithm acts as an explorer, identifying underlying mathematical clusters, cohorts, and groupings that humans cannot manually see.

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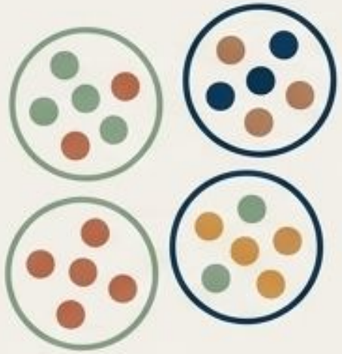
The Four Pillars of Unsupervised Learning

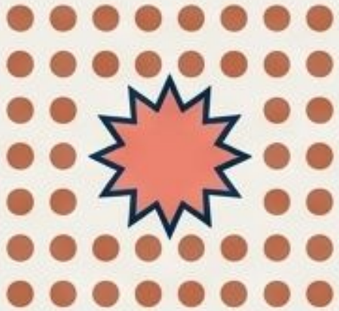


Clustering

Grouping similar multi-dimensional data points into distinct categories based on shared behaviors.

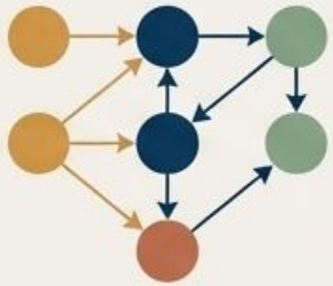
Dimensionality Reduction

Compressing thousands of variables into their most essential features without losing meaning.



Anomaly Detection

Identifying rare, unexpected events that deviate wildly from the established norm.



Association Rule Learning

Discovering hidden "if-then" relationships between seemingly unrelated items.

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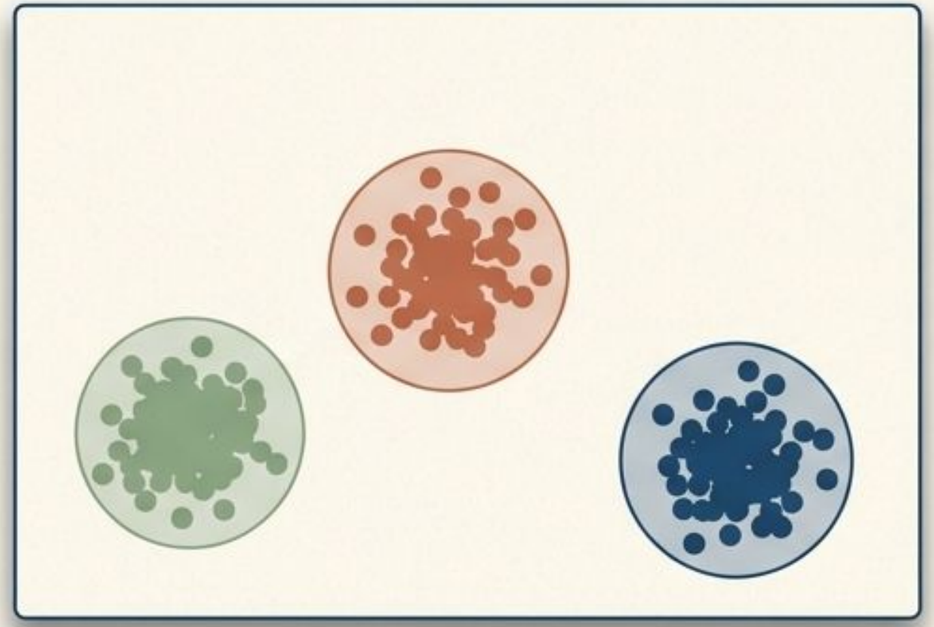
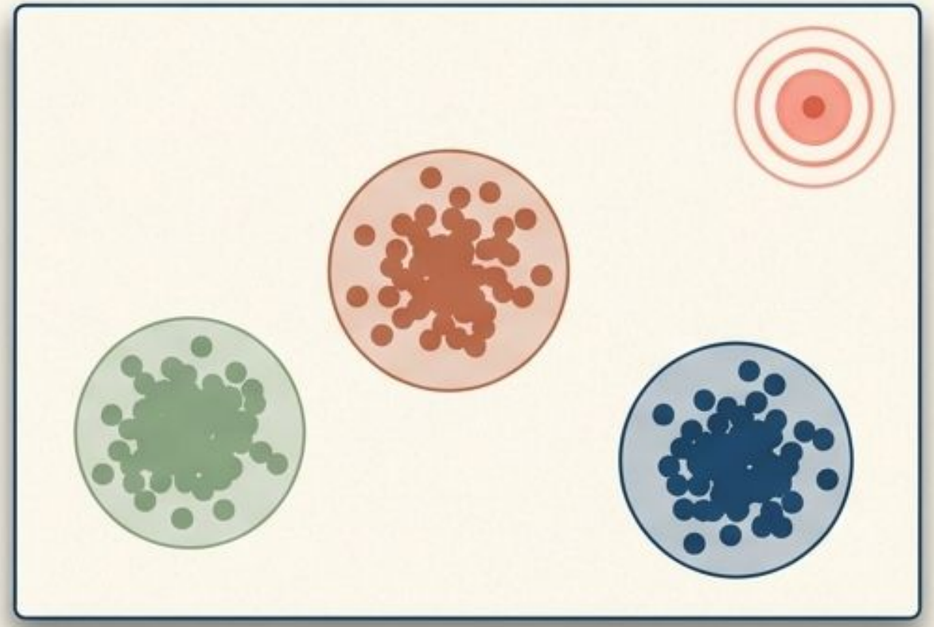
Defining the Norm and Spotting the Outlier

Step 1: Establishing the Norm (Clustering)



Algorithms segment multi-dimensional data into logical cohorts to understand standard operating behavior.

Step 2: Isolating the Abnormal (Anomaly Detection)



Once the baseline norm is established, catastrophic outliers—like manufacturing defects or credit card fraud—are instantly flagged.

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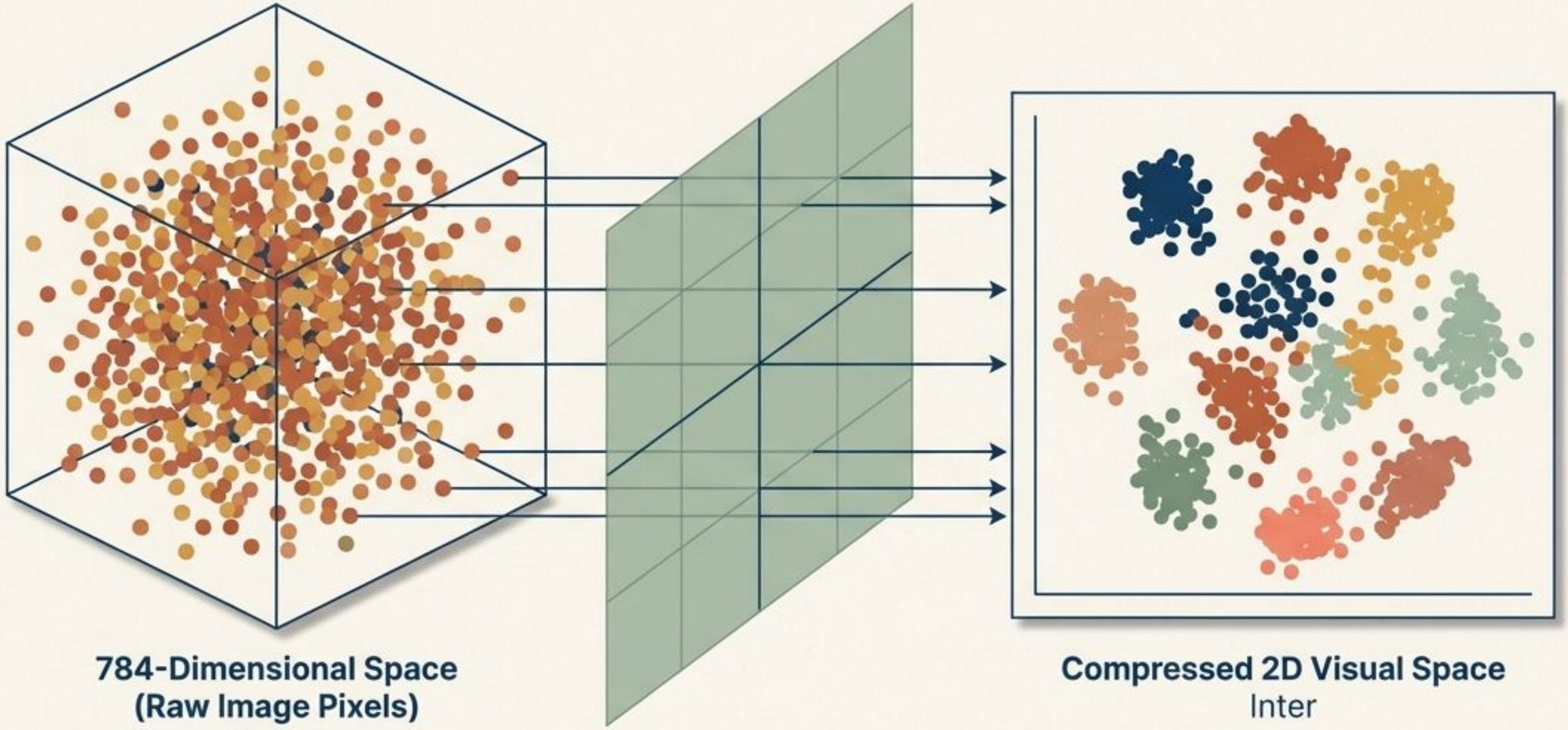
Dimensionality Reduction: Signal Through the Noise



784-Dimensional Space
(Raw Image Pixels)
Inter

Compressed 2D Visual Space
Inter

Algorithms strip away redundant dimensions while preserving core relationships. By compressing 784 variables down to just 2, we enable human visualization of complex data (like handwritten MNIST digits) while drastically reducing computational load.

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Association Rules: The Walmart Case Study

Mining the Market Basket

When Walmart analyzed millions of unorganized shopping receipts, the algorithm uncovered a hidden correlation: customers buying baby diapers frequently bought beer at the same time. By placing these physically unrelated items next to each other, sales spiked.

Association learning mines vast datasets to find hidden ‘If X, then Y’ purchasing patterns.

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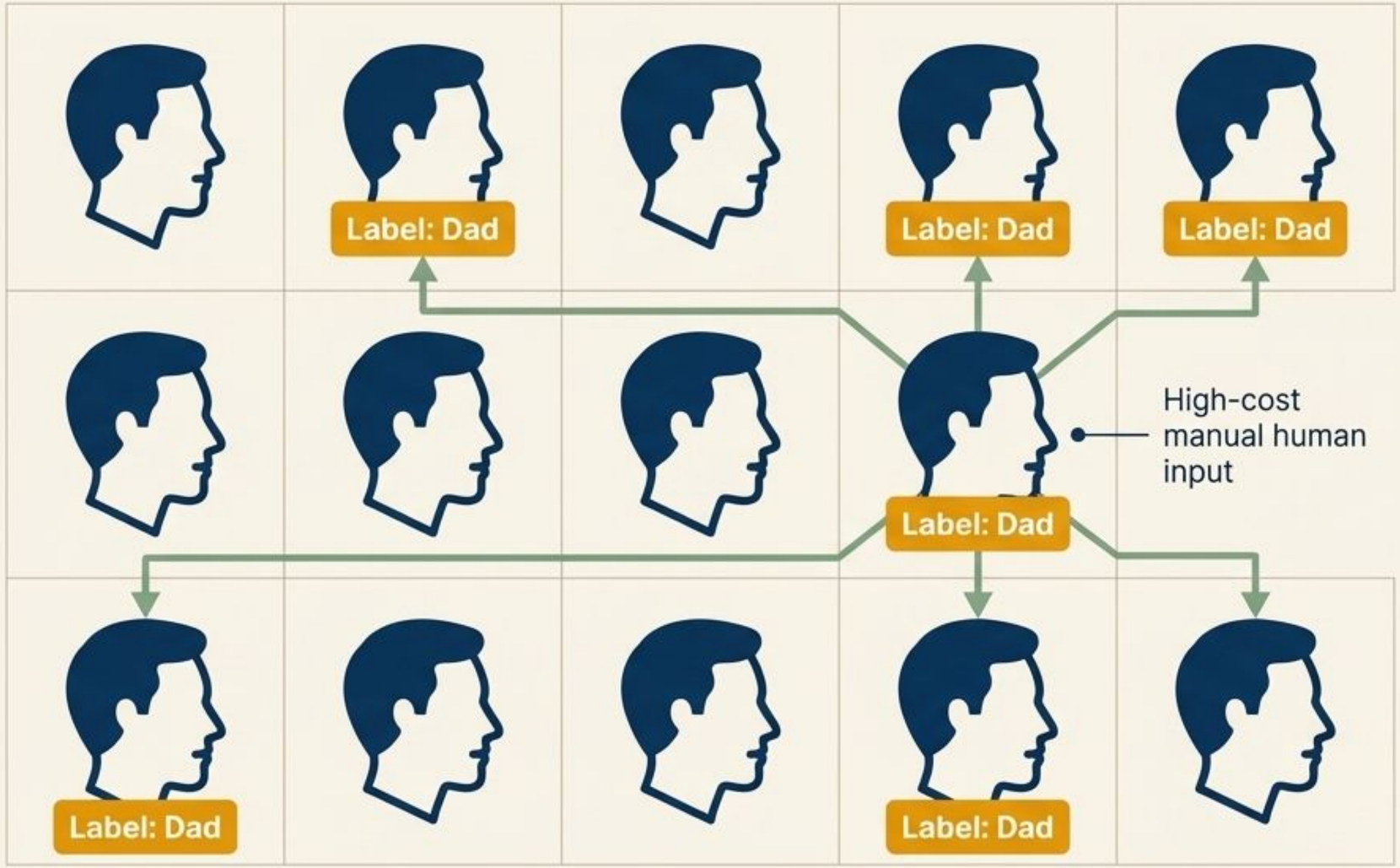
Semi-Supervised Learning: Scaling Human Effort



The Google Photos Paradigm

Manually labeling millions of photos is impossible. By asking a user to identify just one face, the algorithm uses unsupervised clustering to find similar facial structures, then propagates the supervised label to the entire cluster instantly.

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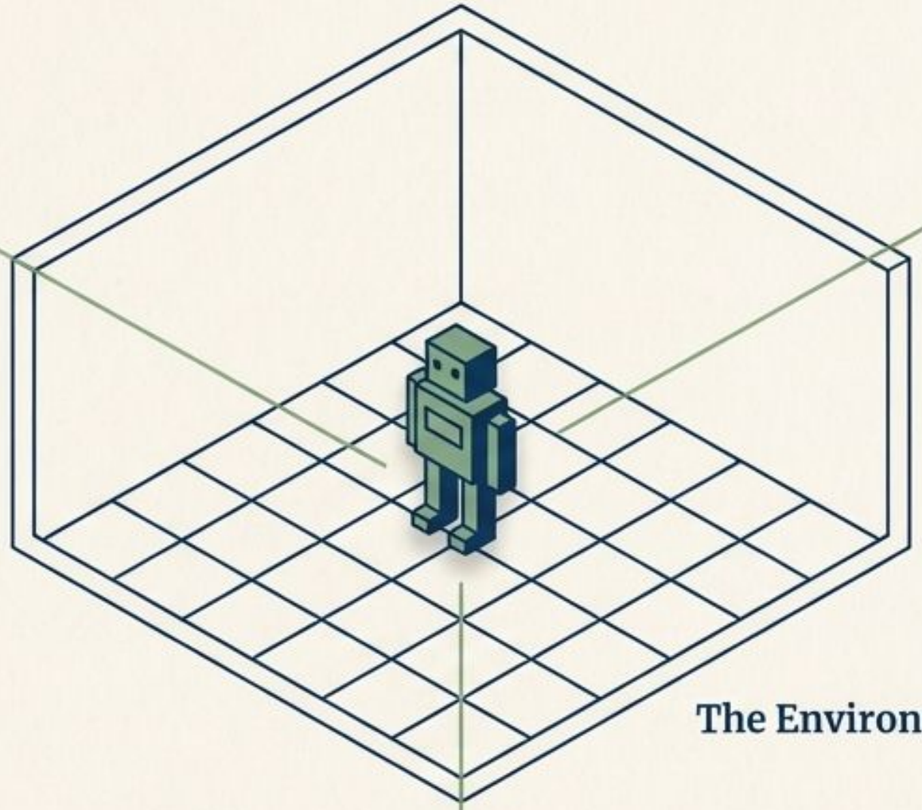
Reinforcement Learning: The Zero-Data Paradigm

Blank Slate

Supervised and Unsupervised models learn from static, historical data. Reinforcement agents start with zero data.

Dynamic Interaction

The agent learns purely through live trial, error, and interaction within its boundaries.



The Environment

DeepMind's AlphaGo

This architecture enables mastery of highly complex systems—from autonomous driving to defeating world champions in the game of Go.

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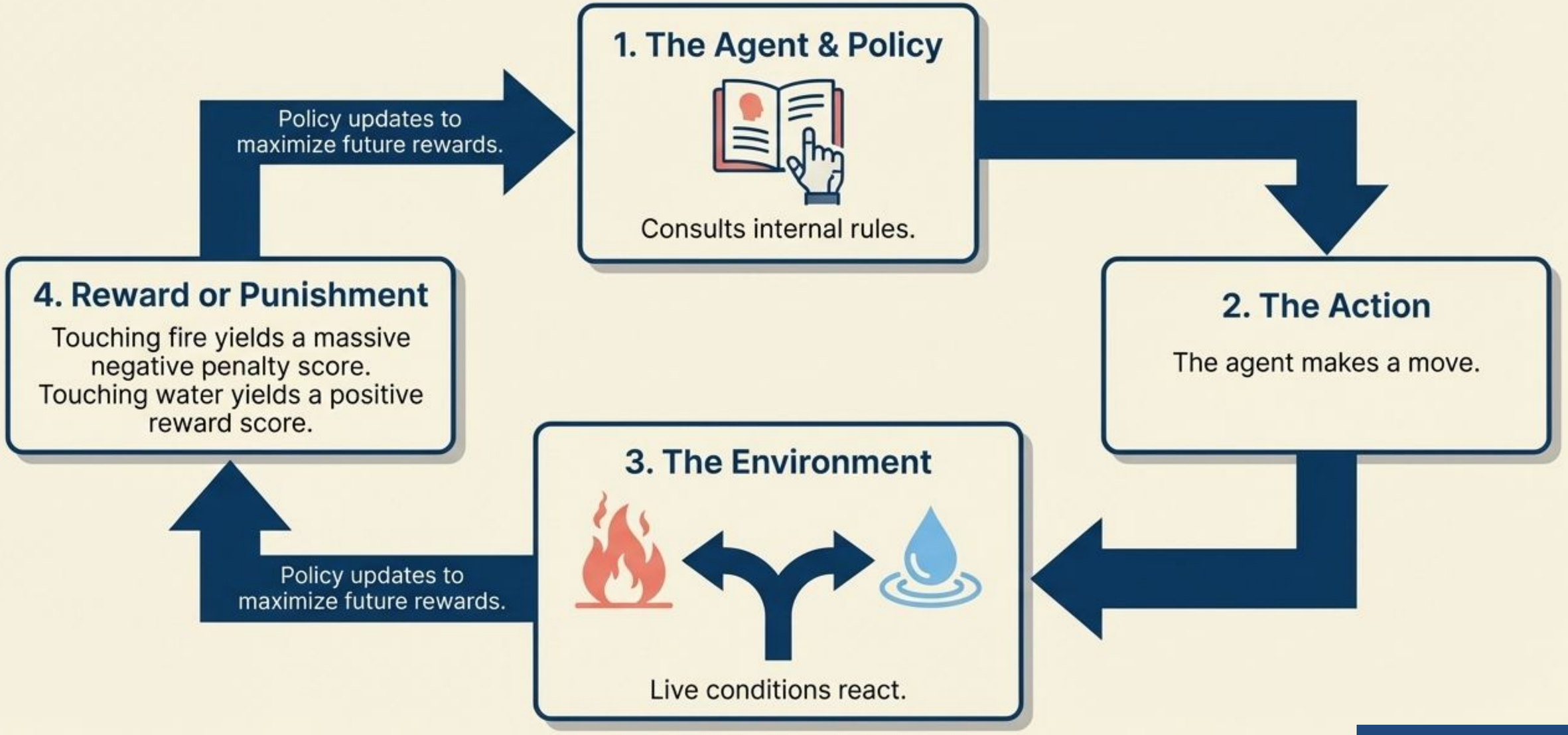
The Trial-and-Error Feedback Loop



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The Machine Learning Master Taxonomy

Algorithm Category	Data Requirement	Human Effort	Core Objective	Classic Application
Supervised	Fully Labeled (Inputs + Outputs)	Very High	Predict specific future outcomes	Predicting graduate salaries
Semi-Supervised	Hybrid (Mostly Unlabeled)	Low	Auto-label data at massive scale	Grouping faces in Google Photos
Unsupervised	Completely Unlabeled	Zero	Discover hidden structures and norms	Walmart market basket optimization
Reinforcement	No Historical Data (Live Environment)	Zero	Maximize rewards via strategic actions	Training an AI to win at Go

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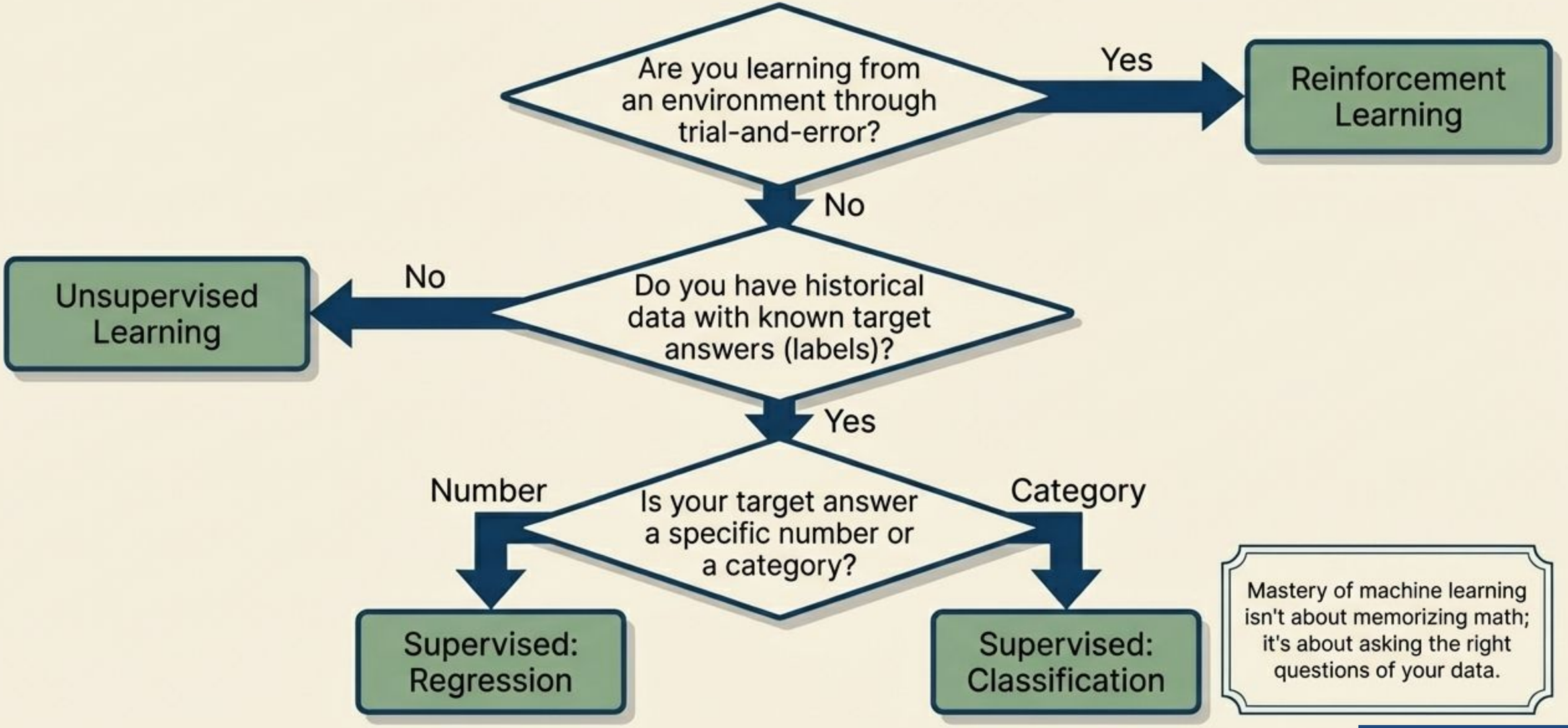
The Algorithm Decision Tree



Mastery of machine learning isn't about memorizing math; it's about asking the right questions of your data.

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